EXASOL EXATHON · TRACK B · MONEY, RISK & FINANCIAL DERIVATIVES

# LiqShield

## Will we survive Friday’s margin calls?

Collateral & liquidity stress intelligence: eligible collateral, shortfall timing and funding cost, computed in SQL.

[Team name] · October 2026

# Headline collateral overstates what you can actually pay with

### Encumbered

Assets already pledged elsewhere cannot be posted again. In our book, $8M of AAA bonds is locked.

### Haircuts

Collateral is worth less than market value: 0% on cash, 2% on Treasuries, 25% on equities.

### Settlement lag

Bonds take a day to become cash, equities two. A call on T+1 cannot use T+2 money.

### Late inflows

Incoming settlements slip while margin calls grow. Stress hits all of it at once.

**$32.0M → $22.55M** on the books versus what can actually be mobilised

# One question, four numbers

Can we meet time-sensitive obligations with the cash and collateral we can actually mobilise?

**$22.55M**

Eligible collateral after haircuts and encumbrances

**-$4.49M**

Shortfall severity: the deepest buffer breach under severe stress

**T+4**

Time to shortfall: the first day the buffer goes negative

**≈ $1.2k**

Intervention cost: the cheapest in-time funding that closes the gap

# Four SQL steps, no application maths

1

## Eligibility

Join collateral to haircut rules, drop encumbered assets, apply the stress haircut.

2

## Time ladder

Bucket cashflows T+0 to T+7, shift inflows, scale margin calls, window-sum the buffer.

3

## Shortfall metrics

Severity, first breach day and days below zero in a single pass.

4

## Funding waterfall

Rank facilities by cost, exclude those too slow, draw until the gap is closed.

The core of the ladder, one window function:

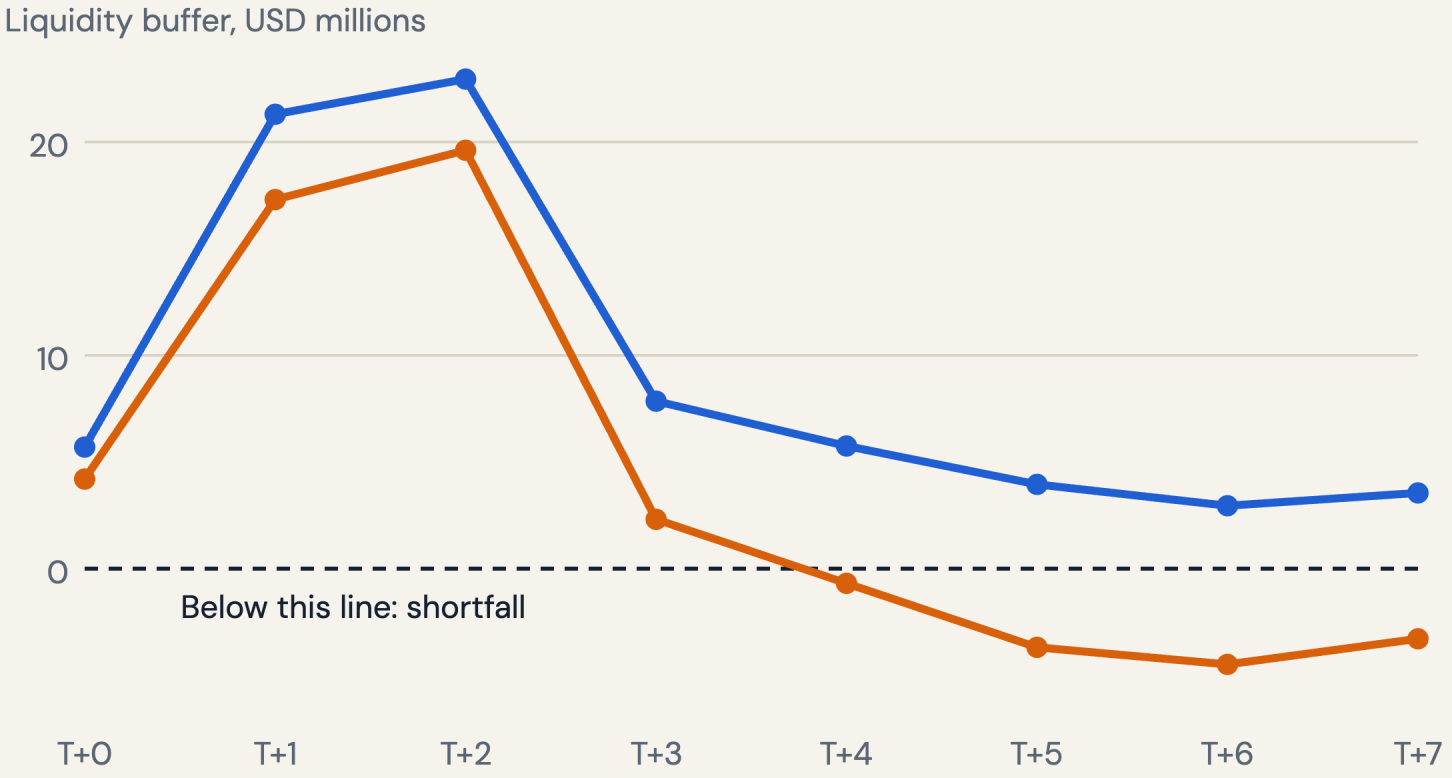
```
MOBILISABLE + SUM(NET_CASHFLOW) OVER (ORDER BY DAY_OFFSET) AS LIQUIDITY_BUFFER
```

# Same book, three scenarios

| Scenario | Haircut | Delay | Margin calls | Lowest buffer | First breach | Peak gap |
|---|---|---|---|---|---|---|
| Base | +0 pp | 0 days | ×1.00 | $2.95M | none | none |
| Moderate | +5 pp | 1 day | ×1.10 | −$0.27M | T+5 | $0.27M |
| Severe | +10 pp | 2 days | ×1.20 | −$4.49M | T+4 | $4.49M |

Synthetic data, as of 2 Nov 2026. Mobilisable collateral is $22.55M. The designed stress event is a $15M Bank_B margin call on T+3: survivable in the base case, a breach once haircuts, delays and margin scaling stack up.

# Buffer by day: base vs severe stress




**Base case**

**Severe stress**

Severe stress first goes below zero on T+4 and bottoms out at −$4.49M on T+6.

The T+3 margin call is survivable in the base case. Add +10 pp haircuts, late inflows and calls ×1.2, and it is not.

# Funding actions: cheapest first, only if in time

| Facility | Limit | Available | Annual cost | Severe stress (breach T+4) |
|---|---|---|---|---|
| Central-bank credit line | $5M | T+0 | 2.50% | Draws $4.49M, cost ≈ $1.2k |
| Committed revolver | $8M | T+1 | 3.20% | Available, not needed |
| Overnight repo line | $10M | T+1 | 4.50% | Available, not needed |
| Equity fire-sale | $3M | T+2 | 12.00% | Available, not needed |
| Parent support (uncommitted) | $20M | T+4 | 6.00% | Available, not needed |

Lead time matters: in a tougher case (+20 pp haircuts, 2-day delay, calls ×1.3) the breach moves to T+3, and parent support, available only from T+4, is excluded as too late.

# Every number is checked three ways

1

## Hand calculation

```
5.00 + 9.80 + 3.80 + 1.70
+ 2.25 = $22.55M eligible
```

Cash, Treasuries, AAA, BBB and equities after haircuts, with the pledged $8M excluded.

2

## Independent oracle

A pure-Python re-implementation shares no code with the SQL.

11 pytest tests compare the two across four scenarios, plus hand-calculated constants.

3

## Excel model

A formula-driven workbook recomputes the ladder from first principles.

Switch scenario and the PASS cell confirms a match to the cent, all 8 buckets.

# Built for Exasol, honest about status

## Designed to run in the database

- Joins, filters and window-function ladders in SQL
- Funding waterfall using running window sums
- Python UDF for Monte Carlo haircut shocks, inside the database
- pyexasol connector, schema, loader script included

## Where we are today

- Verified end to end on DuckDB, running the identical SQL
- 11 of 11 tests pass; Excel shows PASS in all three scenarios
- Our local Exasol Personal deployment would not start on submission night
- The Exasol path is implemented but not yet verified on a live instance

# Next: run it on Exasol at scale

- Verify the Exasol path and the in-database UDF on a live Exasol Personal deployment
- Scale to millions of positions and intraday time buckets
- Add multi-currency, netting sets and counterparty limits
- Connect live collateral and cashflow feeds

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Code: github.com/krithiyaragarla/liq_shield_exasol

Demo video: [video link]

Synthetic data (CC0) · MIT-licensed code